

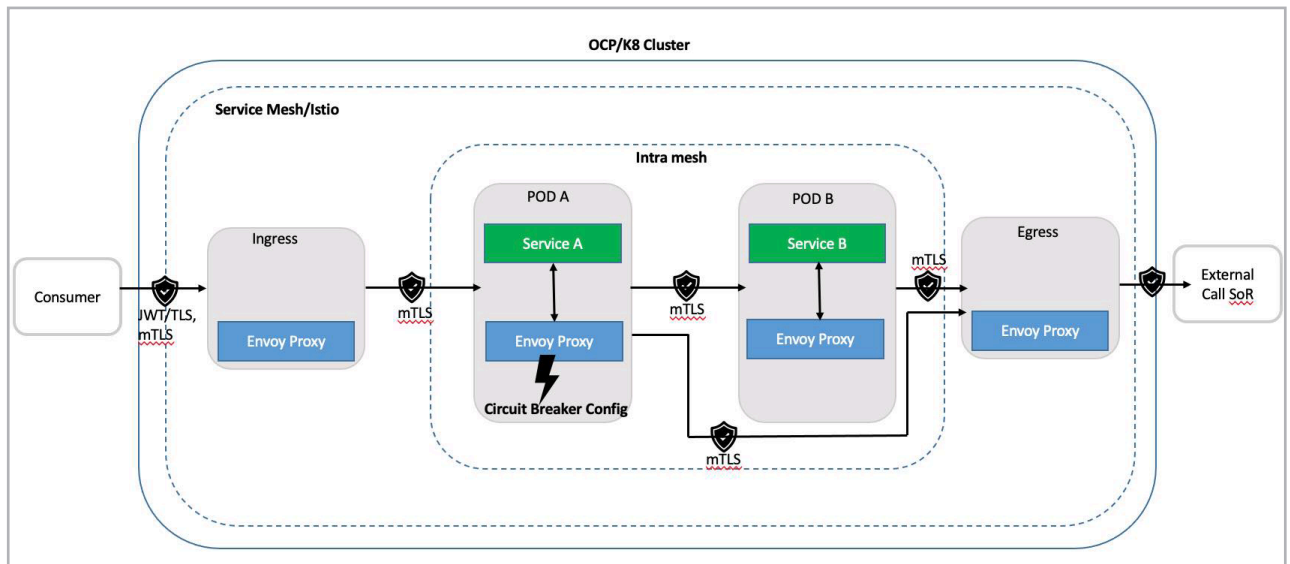


Figure 2: Circuit breaker between services in service mesh

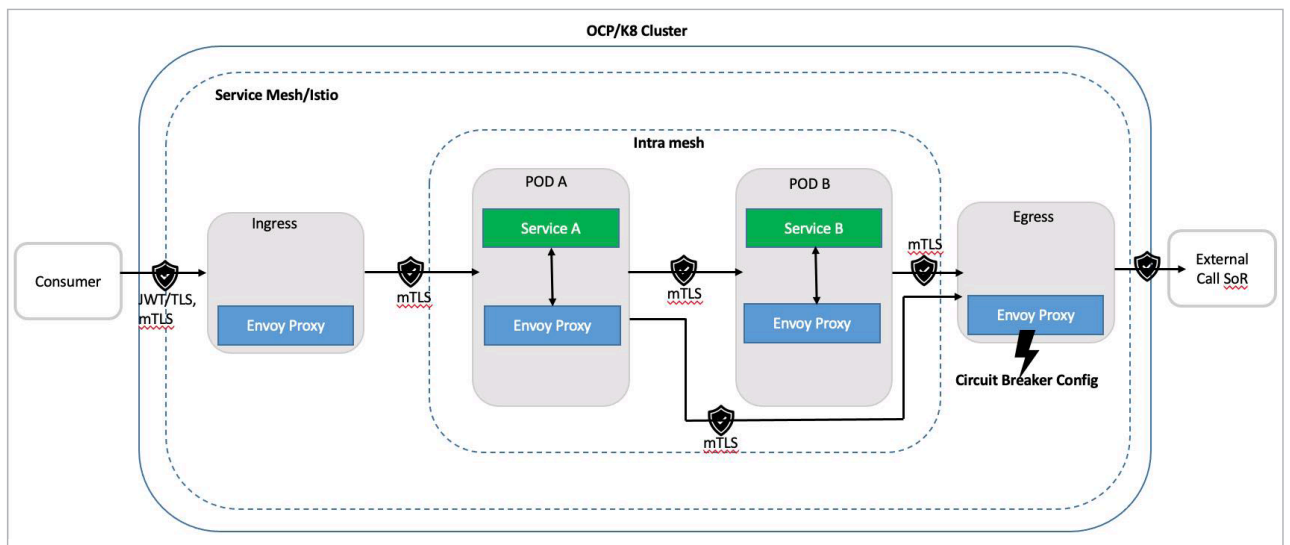


Figure 3: Circuit breaker at the point of egress

- iii. *Envoy proxy* defining a rate limit service, as it does not tell by itself whether a request has to be passed through or rate limited.

Application and impact of the pattern

- This pattern is produced with reference to Istio 1.8; however, it is applicable for microservices deployment in DIY Kubernetes or OpenShift Container Platform 4.6.
- Applicability of the specific pattern described here depends on individual application design. It must consider the effects of when a circuit breaker opens based on functional and operational requirements. Consideration should be given to the applicability of a combination of

resiliency capabilities to manage the flow and content of requests to a service as well as the behaviours where a problem with a service is detected.

Securing services within a service mesh

The problem it solves

In distributed microservices architecture, authentication and authorisation of the service consumers and security of payloads between services are needed. However, configuring or coding this for every service/component can become significantly complex and require much effort to manage as the number of services/components grows, since the integration touchpoints of the services also